

Insight generation from business analytics

There are three primary characteristics of business analytics required for it to be successful. It should generate values and to do so, it must:

1. be purposeful
2. deliver new insights
3. be dynamic and actionable.

According to Stubbs (2011), 'Business analytics should go beyond plain analytics, requiring a clear relevancy to business, a resulting insight that will be implementable and actionable, and performance and value measurement capabilities to ensure a successful business result'.

To be purposeful, business analytics must:

- align with business functions
- align with the knowledge of the person taking action
- address management objectives and issues (strategy, performance, compliance, risk)
- derive real value from the ideas generated through conversations by closely investigating information.

To be insightful, business analytics must:

- discover new factors or information
- create awareness of facts previously unknown
- surface cause–effect information
- enable future–looking decision making.

And finally, to be dynamic and actionable, business analytics must:

- support decision making and action taking
- improve discovery and insight, determination and resolve, innovation and creativity
- lead to actions that integrate well with organisational processes
- initiate actions that improve or generate revenue.

Data, information and insights

Before you begin the process of analytics, it is important to understand what data, information and insights stand for.

Figure 2.1

Data, information and insights



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[\[Image transcript\]](#)

- **Data:** Data is referred to as raw and unprocessed facts that are usually in the form of numbers and text such as quantitative (measured) or qualitative (observed). In the image above, you will notice that data is the largest section as it contains a lot of information. In business analytics, not all data is useful and useful data must be derived from this raw data.
- **Information:** This useful data that is processed, aggregated and organised into a more human–friendly format that provides more context in the form of data visualisations, reports and dashboards is called information. In the image above, you will notice this is a must smaller section than data as this information has been derived from data.

- **Insight:** Not all information is useful to make strategic business decisions. The information is further analysed to generate insights. Insights are generated by analysing information and drawing conclusions. Out of all the insights, the actionable insights as shown in the right pyramid should be used to influence decisions and drive changes.

Characteristics of actionable insights

For a business, identifying actionable insights is the main purpose of implementing business analytics. Therefore, it is important to understand the key characteristics of actionable insights so that they can be identified and implemented.

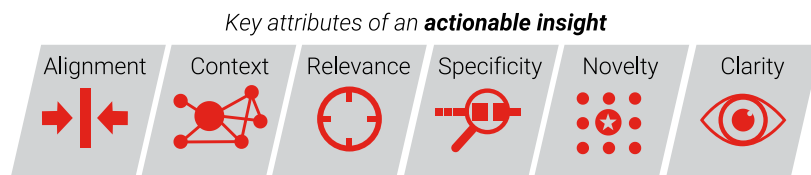
Actionable insights

Actionable insights are the subset of findings from an analytics exercise that are useful for decision making and generate value for an organisation. Data is transformed by an analytics technique into findings (also known as outputs or results) and some of these findings will be common knowledge or even common sense, but some other findings will be novel and generate decision value for the individual or organisation. Such findings are actionable, thereby we use the term 'actionable insights'.

The key attributes of actionable insights are represented in the image below.

Figure 2.2

Key attributes of actionable insights



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Let's discuss each attribute in detail.

Click on the below headings to reveal further information.

✓ Alignment

- An actionable insight must be:
 - in line with business goals and/or strategic initiatives within organisations. Only when the insight is aligned with a business goal is it more likely to be valuable and drive actions.
 - linked to KPIs or any other key metrics of an organisation that are used to measure the performance of the business. When insights are aligned with KPIs they are more likely to drive actions.
 - connected to a lever the business can act on.

> Context

> Relevance

> Specificity

> Novelty

> Clarity



Read

Read the following article exploring data, information, insights and actionable insights in detail. This reading will strengthen your foundational knowledge of business analytics and will be useful in future weeks and to complete Assessment 1 — online quiz.

[*Actionable insights: The missing link between data and business value*](#) (Dykes, 2016)

Applicable areas of analytics and challenges

Areas of application

Some of the areas within an organisation where business analytics can be applied are shown in the table below. You will explore these in detail in future weeks.

Table 2.1

Areas of application

Areas		Questions that you'd like to answer using business analytics
1.	New customer acquisition	Which residents in a postcode should receive a coupon in the mail for a new store location?
2.	Customer loyalty	What advertising strategy best elicits positive sentiment toward the brand?
3.	Cross-sell / up-sell	What is the best next product for this customer? What other products is this customer likely to purchase?
4.	Supply optimisation	How many 75-inch OLED TVs should be in stock? (Too many is expensive; too few is loss of revenue.)
5.	Financial forecasting	What weekly revenue increase can be expected after the Valentine's Day sale?
6.	Product placement	Will oatmeal sell better near granola bars or near baby food?
7.	Churn analysis	Which customers are most likely to switch to a different wireless provider in the next six months?
8.	Fraud detection	How can I identify a fraudulent transaction?

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Challenges

Business analytics comes with its fair share of challenges. These are:

- Quality of data — one of the biggest challenges of big data is around the veracity of the data. To ensure that the data is of good quality, a lot of cleansing is required. In the analytics industry, it is often said that analysts spend almost 80 per cent of their time cleaning the data and 20 per cent performing analysis on the clean data. In recent times, there has been a growth in the hiring of data engineers. The primary responsibility of a data engineer is to get the data ready for analysis by data scientists.
- Data is more unstructured these days (versus structured data) — not all data that becomes available is structured. Data from social media platforms is often unstructured (in the form of images and videos) and requires a lot of structuring before analysis can be performed.
- Volume of data is growing very fast (with the surge of technological devices and platforms, IoT) — real-time data and the growth in the number of connected devices being used by an individual and analysis being performed are at an uneven pace.
- Poor business processes (garbage in equals garbage out) — if an organisation inputs useless data, the output will not be useful.
- Privacy (trade-off between utility and privacy) — this is an ongoing conflict. Where do you draw the line, and often a trade-off is required.



Watch

Industry insights with Manpreet Kaur

Watch this video [15.54 min.] to hear from Manpreet Kaur, Consultant with Infonyx, as she discusses real-world applications of R and Python. Hearing from industry professionals helps bridge the gap between theory and practice, providing up-to-date information on current trends, and offering insights into professional skills and expectations. These industry insights will enhance your learning through real-world applications and exposure to diverse career paths.

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[\[Video transcript\]](#)

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